

CHAPTER FOUR

Monitoring Your Training

It should be self-evident that the higher your goals, the harder you will have to work to achieve them. In doing this hard work, you will be pushing perilously close to the boundary of physical and mental stress you are capable of handling before something goes wrong. Most athletes look for some method of monitoring and analyzing their training load and/or the fatigue arising from that load. While a few rare individuals seem to manage using only their intuition and can get by with no monitoring tools, most of us will benefit greatly from learning to use one or more tools.

High is a relative term. An intolerably high training load for a beginner won't even be a maintenance load for a world-class athlete. There is no one-size-fits-all training prescription. *This is why we've created a book that teaches you the theory and practice of training rather than doling out a set prescription which may or may not fit your needs.* Thoughtful application of sound training principles accompanied by attentive monitoring always results in the best outcome.

First and foremost, adhere to the principle of gradualness in the progression of your training load. Except for rare instances of premeditated overreaching, give yourself a chance to absorb the previous load and build progressively. Our bodies do not like drastic changes.

Take an active interest in understanding the effects that your training is having on your body. This involves learning to listen to and interpret the subtle signals your body is sending. Listen carefully while they are still being whispered and before they become shouted alarms.

Ways to Monitor for Recovery

It is only natural that when you are exploring the limits of what your body can handle, you may occasionally overstep. Short-term tiredness goes with the

territory. Normally fatigue is short term because you recognize it early and deal with it by reducing the training load for a day or so, which is almost always sufficient to restore your body's homeostasis and get you back on track again. Pay attention to this short-term fatigue and don't let it become chronic or you risk sliding into the realm of overtraining.

Don't train compulsively. If you are motivated enough to be reading this book, in all likelihood when you lack the motivation to train it is because your body is telling you to back off in the only way it can. Days on which you lack motivation to train are a clear warning sign.

Just because the training plan calls for a hard workout does not mean that you will be ready to execute it effectively or more importantly, benefit from it. Fatigue has a determining effect on your preparedness to train. Accurately assessing your state of fatigue is critical to effective training. Here's a quick look at some simple ways you can monitor your recovery.

Fatigue

Endurance athletes must learn to embrace fatigue. We train to increase our resistance to it and we race to see how well we can tolerate the pain and discomfort of it. Fatigue, in one form or another, is the stimulus that makes you fitter and it is by necessity the nearly constant companion of endurance athletes. Staying on the good side of the fatigue curve is one of the most important factors in successful training. Increase the training load faster than your body can adapt to it and fatigue can very quickly go from being your friend to your worst enemy. Knowing fatigue in all its guises and how to control it so it does not control you is a learning process. Some athletes never figure it out and wonder why they come up short on the big day. But most of us will benefit from applying a little structure and a few rules to help us learn from studying fatigue and how it affects our individual body.

NOTES FROM KÍLIAN

Listen to your body. Sometimes it will say you can do it and you can push; other times you realize you need to stop before the prescribed time is over and say the training is enough for your body on that day. You must know how far you can go before feeling overtrained. The early years of training are best spent learning about your body and when you can push, and when not to push.

You want to discover your system and we each need to learn to recognize these feelings in our own bodies. Sometimes you may become tired and not feel good after a lot of training days; that's okay because it's all part of learning about yourself. Make a note; remember (and study) what you did that led to that feeling. And move forward.



Caleb Ambrose relieves some fatigue at the twenty-mile mark during the thirty-four-mile circumnavigation of Mount St. Helens by the Loowit Trail, Washington. Photo: Steven Gnam

Understanding fatigue in its various guises is important if you want to manage it productively. We can categorize fatigue into three general groups. These types of fatigue often overlap and interact with one another, but for simplicity's sake we'll look at them individually.

1. **Calcium Accumulation:** When calcium ions accumulate in the muscle cells, muscle contractions are inhibited. The calcium pump that moves these ions back into the intercellular space fails to keep up with the build-up and very soon the muscle loses its ability to contract. This effect is most noticeable in short-duration, high-strength contractions. Think of the fatigue you experience when doing pull-ups. The duration of the event resulting in this sort of fatigue is usually under fifteen seconds. Recovery from this is on the scale of a few minutes. Typically, this type of fatigue is self-limiting as your ability to contract those muscles powerfully will naturally end that type of movement; you can manage one more pull-up. However, if you drive yourself into this state of fatigue many times in a workout (as with several sets to failure of pull-ups), the following forms of fatigue can be induced and recovery can take days.

2. **Metabolic:** On a short time scale, there are products of metabolism that accumulate in the muscles that inhibit the synthesis of ATP. Lactate is the most well-known, but the increased acidity from the accumulation of hydrogen ions is another one. This reduction in ATP production manifests as a reduction in muscle power and a consequent slowing. Think about the fatigue you experience when sprinting up a hill for sixty seconds at near-maximum effort. This fatigue is often associated with a burning feeling in the muscles. Recovery from it is on the scale of a few minutes. On a longer time scale, the glycogen stores available to power the metabolic process of ATP production can become depleted and muscle power drops as a consequence. Think of the classic “bonk” during multi-hour events that often is accompanied by mental confusion and a lack of coordination. Recovery from this is on the scale of a few hours to a few days.
3. **Neurologic:** The electrical signals transmitted from the brain’s motor cortex, along the motor nerves to the muscles, require neurotransmitter chemicals at each synapse to relay the signal along its way to the muscle. When these chemicals become depleted through prolonged high firing rates, the electrical impulse becomes weakened and muscle power drops. Repeated or longer-duration, exhausting efforts of the first two types in this list can result in this form of fatigue. Normal recovery from this is on a scale of a few hours to several days.

THE CENTRAL GOVERNOR

A credible theory about the limits of human performance was proposed by the grandfather of exercise science, A.V. Hill, in the early twentieth century. Hill’s VO_2 max theory has been modified and strengthened over the years. In Section One we mentioned this other highly credible theory about fatigue and the limits of human performance: Dr. Tim Noakes believes that the brain has a built-in self-preservation instinct that limits how far out of our homeostasis we can get before alarm bells begin to sound. Interestingly, while the causes for each of the foregoing types of fatigue are different, the result is the same: The brain reduces the muscle power output until homeostasis is restored. Noakes and many others have provided ample evidence to make this the most robust theory of fatigue to date. What does this mean for you? Noakes proposes that frequently pushing the body out of its comfort zone (away from its homeostasis) makes the governor more comfortable in allowing ever-increasing divergence from homeostasis before it begins to shut things down. You do this when you train. And it seems very likely that fatigue in all forms depends upon this, as yet, poorly understood relationship between the muscles and the brain.



Ryan Sandes rolls it out in the dormitory of Lac Blanc Mountain Hut, Chamonix, France.
Photo: Kelvin Trautman

The Fatigue Matrix

The next step after grasping the causes of fatigue is to look at how the responses manifest so that you can recognize them. Recognizing the causes and results then leads to action: what you can do to mitigate the fatigue in order to speed your recovery. The suggestions we provide below are based on our accumulated experience and study. First, consider that we can group fatigue symptoms into two rough types: Group A: Muscular (and nervous system), Group B: Energy.

As with most of the other simple models provided in the book, this one is not 100 percent accurate. Fatigue causes overlap and are somewhat interdependent. You may have symptoms of all the following. Keeping these general categories in mind can help you decide on the best course of action.

Group A: Muscular

Are your muscles sore or are do they feel dead, heavy, and flat? In the first instance of soreness, this is caused by micro-tears in the muscles. In the second case, of dead legs, the neurological fatigue mentioned above or glycogen depletion are the primary culprits.



Dirt, grit, blisters, sunburn, and sweat pustules are nothing compared to the relief and satisfaction Jenn Shelton finds after completing her long-term goal of running all 210 miles of the John Muir Trail. It took her four days to traverse the backbone of the Sierra Nevada from the high point of Mount Whitney (14,505 ft) to Yosemite Valley, California. Photo: Ken Etzel

Sore Muscles

Did you increase the intensity of your strength workout? Did you do some faster or longer downhill running? These are the kinds of things that might leave your muscles sore to the touch and make descending a set of stairs painful. The common term for this is Delayed Onset of Muscle Soreness (DOMS). It usually peaks forty-eight hours after the workout. Mostly you just have to tough it out and do light aerobic exercise during the DOMS period. Here are the recommended do's and don'ts to deal with this condition.

Do:

Light aerobic recovery workouts

Swim, or at least lightly kick in the pool

E-stim on the recovery setting

Light massage

Ibuprofen after twelve hours post workout

Ice bath after twelve hours post workout

Frequent light stretching to reduce muscle stiffness

Get extra sleep

Eat extra protein, especially before bed
Self-maintenance rolling with roller or ball

Don't do:

Speed, power, or strength work
High-intensity aerobic interval work
Overstretch

Dead, Heavy, or Flat-Feeling Leg Muscles

This condition is usually caused by increasing training load too much in too short of a time. Have you increased the distance, vertical, or the intensity of your training in the past ten days? Do you notice a lack of spring in your running stride? Do your legs feel extra fatigued when you're just climbing a set of stairs? Do your legs ache at night in bed? Has your performance or pace dropped off and do your workouts seem to take more effort? All these are big red flags and if ignored, can lead to a state of overtraining (covered in detail in a few pages). This is a complex type of fatigue that is probably a combination of Group B: Energy (glycogen depletion) and neurological fatigue. If your legs are not refreshed after a couple of very light days, then you need even more of the following.

Do:

Extensive use of the pool for swimming/light kicking exercise
Self-maintenance rolling with roller or ball (may need twice daily)
Deep-tissue massage
Compression garments
Elevate legs above heart
Increase post-workout carb intake

Ferritin

Ferritin is a very important protein in your blood that is responsible for storing and transporting iron across the cell membrane into the hemoglobin molecule. Elemental iron can't cross that barrier without being attached to ferritin. Since iron binds with oxygen in the hemoglobin, when ferritin levels are low, the oxygen-carrying capacity of your blood will be reduced. For endurance athletes, this results in a performance decline. The normal ranges of acceptable ferritin levels do not consider

the demands of athletes, so can't be relied upon. The standard for "low" ferritin reading is 12ng/ml. For endurance athletes, a ferritin level below 50ng/ml will result in some reduced endurance performance. Levels below 30ng/ml will show as dramatically reduced performance and chronic fatigue. It is worth checking ferritin levels with a simple blood test if you have been experiencing these symptoms. It is often overlooked by physicians who are not used to working with athletes.

Light recovery effort aerobic cross-training: cycling, rowing
Increase quality and quantity of sleep

Don't do:

Extensive aerobic workouts, especially with much vertical
Intensive workouts such as Zones 3–5 interval sessions
Muscular endurance workouts
Anaerobic endurance workouts like extended hill sprints
Drink much alcohol

Group B: Energy

If you are reading this book, in all likelihood you love to train. You don't need someone pushing you out the door to do your workouts. In fact, you probably need someone to hold you back from time to time. Your sport most likely helps define you and when you feel lethargic and unmotivated, this can create emotional stress. But, when you feel unmotivated and with low energy, you should take that as a warning sign. Your body is sending you a message and you may be suffering from this type of fatigue. Are you waking up tired in the morning? Even when sleeping more? Needing more sleep in the day? Are you feeling flat? Are you avoiding friends? Is work or family life more stressful? Then consider following this course of action for a few days to see if you feel some improvement.

Do:

Increase quality of your food
Increase quantity and quality of sleep
Make more social time with friends
Consider blood tests for cortisol, ferritin, testosterone, mononucleosis
Simple heart rate test; resting or stair test (see below)
Take a fish oil supplement
Stay off computer and/or similar devices one hour before bed
Easy and short Zone 1 aerobic workouts (can include a few short accelerations)

Don't do:

Intensive workouts such as muscular endurance or Zones 3–5 interval sessions
Consume alcohol
Add stress to your life, either physical or psychological

Perception

How you feel when you get ready to train or after a warm-up is the old standby and one we all use intuitively. Even if you do use one of the more concrete metrics listed below, you still need to do a reality check and confirm that your body's



Filippo Beccari going all out during a skimo competition in Italy. Photo: Federico Modica

sensations agree with the numbers. An important function of a warm-up is to assess your readiness to train. Be prepared to reduce or even cancel the workout based on how you feel.

Resting Heart Rate

An elevated resting heart rate suggests that your body's sympathetic nervous system is kicked into high gear, possibly as a result of overstimulation from a big training load. It can also be due to an infection or impending sickness, poor sleep, or some other stress in your life. But whatever the cause, you need to be watchful and monitor your response to training more closely over the next few days to see if your resting heart rate deviates even more or if it begins to return to a more normal level.

A simple morning heart rate test is worth trying out for yourself and seeing if you can connect cause and effect. Most athletes' results will be spotty with this resting heart rate method. There are other factors that affect the resting heart rate besides fatigue or illness. This is not to say that you should disregard an elevated morning pulse, but it should be weighed with other factors. Be sure to take the test at the same time and under similar conditions each day.

Step Test

This very simple test can be used to help gauge your recovery state/preparedness to train. It lightly stresses your nervous system and you simply note the results. Wearing your heart rate monitor, step up and down at a moderate pace onto a stair step. Do this for a minute or two at a fast enough pace to raise your heart rate about 50 to 70 beats/minute. The effort need not be very intense. Just enough to cause a reaction by the sympathetic nervous system. Note the maximum heart rate achieved and then sit down immediately and note your heart rate after one minute. The exact times you use and the stepping rate are not critical as long as you repeat them each test. By doing this test frequently as well as noting the quality of the corresponding training, you will soon see a correlation between the drop in heart rate and your recovery state.

Stair Test

This simple assessment consists of you walking or bounding (taking two steps at once) up a set of stairs that you climb daily and then checking how your legs feel at the top. You'll quickly begin to recognize the feeling that allows you to have a good workout and the feeling that tells you that you need more recovery.

NOTES FROM KÍLIAN

Besides tracking duration at each heart rate, I also track my vertical meters every day.

I note every workout I do on a spreadsheet with a short description of the training and a note about how I felt. I have

recorded all my training this way since I was very young.

If a race goes really well, or if I did not feel fast, I like to go back and see what was before so I can try to understand it.

Use a Training Log to Monitor Training Load

Training effect indicates, in a qualitative way, how your body reacts and adapts to the various forms of training stimuli. *Training load* is a term used to describe, in a quasi-quantitative way, the type and amount of stress imposed by a single workout or period of training: the training load causes the training effect. They both depend on the type, the intensity, and the volume of work done in a training bout. The training load applied in a three-hour Zone 1 run may be roughly equivalent to the training load of a one-hour run containing five one-kilometer intervals done in Zone 4, but the training effect and adaptations will be quite different.

Quantify and Track

Training load is a function of the volume and the intensity of the work. Use a plan to guide you in what workouts to do when, and remember to use a log for recording your workouts. This history will allow you to plan better in the future. The more accurate and informative the recording, the more useful the information will be when you are next planning.

A training log can be as simple as a notebook or as sophisticated as an online logging and planning website. TrainingPeaks.com is the platform we use for our coaching, and the company's proprietary tracking metrics allow for remarkably accurate fitness and fatigue monitoring. Other online services like Strava, MovesCount, Garmin Connect, and Polar Flow also offer training data logging and will all synch to TrainingPeaks, but as of this writing lack its sophisticated tracking capabilities.

TIME

The simplest way to track training load is to keep track of training volume either in hours or miles. Logging miles or kilometers has been the standard metric for road runners for decades. Logging hours is the standard method for mountain athletes. As you know, in the mountains not all miles are created equally. Therefore, hours is a fairly good metric.

ELEVATION GAIN/LOSS

When combined with training time, vertical gain and loss accumulated in a workout or week or month can begin to give an even clearer picture of the training load for the foot-borne athlete. This is because for the mountain runner and ski mountaineer, it is the vertical component of the training that normally includes the intensity aspect of training. Even if the vertical training is done at a low intensity, the amount of muscle mass recruited is more than when training on less steep terrain. Increase vertical and you increase the training load.

How Kilian Records His Training

For running, Kilian counts time, vertical meters, and distance. For skiing, he counts just the time and vertical meters. He also uses a simple scale to describe the technicality of the workout:

Tech 1: On piste ski or good track/easy trail

Tech 2: Off piste or rough track/medium trail

Tech 3: Off piste, some scrambling/off-trail scrambling

Tech 4: Climbing



An unknown runner striving for an A-grade workout in the Lake District of England.
Photo: Jordi Saragossa

While there is no formula that can take into account steepness, footing, size of steps you may be forced to take over rocks, total climb, etc., it's enough to know that a 50-mile (80-kilometer) week with 6,000 feet (1,800 meters) of vertical gain and loss will have less load than a 40-mile (60-kilometer) week with 10,000 feet (3,000 meters). Your legs will let you know which they find more challenging. Record your vertical just as you would time and distance. It is necessary to progressively build the training load during the base-building period; in Section Four of the book ("How to Train") we'll show how much to increase this.

GRADING YOUR WORKOUTS

A useful and simple tool we often use for monitoring training progress is to give each workout a letter grade based on how you feel during the workout and record this value in your log. Using this grading method in conjunction with a recovery-monitoring test will give you some solid feedback.

- A – means you felt like Superman and had plenty in reserve.
- B – means it was a good workout. You completed the task with no problem.
- C – means you did the workout but felt flat or off.
- D – means you could not finish the planned workout or had to reduce it.
- F – means you could not train that day due to fatigue or illness.

By grading each workout this way, you can make a quick scan of your training log over the previous few weeks and see trends or patterns. A good rule of thumb is that if you have more than two Cs in a row, or a C and D within one week, you need to stop and assess what is going on. For some reason, you are not absorbing the training; the load is too high for your current fitness state or the recovery is insufficient.

Not every workout is going to be an A, but if your workouts keep earning poor grades, then your body is trying to tell you something. You may be getting sick and not have any symptoms yet. If you continue without changing something, the Cs will become Ds and begin to fill the pages of your log. At this point, you are no longer adapting to the training and headed toward overtraining.

Returning to Training After a Break

Breaks can be planned or unplanned. Illness, injury, school, work, family, and friends can sometimes thwart your resolve to stick with your plan. Often the best strategy is to accept the minor setbacks with equanimity and try to learn something in hopes of not having to repeat them in the future.

Fatigue

This is perhaps the most common reason for unplanned disruptions in a training plan. See the preceding discussion about fatigue on page 25.

When beginning to train again after a fatigue-related break of more than two days, do not just jump back into training where you left off. It may well be that that load was too high and was responsible for your excessive fatigue. Instead, ease back into training with low-intensity workouts that are reduced in duration by 50 percent for a couple of days. This will give you a chance to assess your actual recovery state. If these go well, then increase to 75 percent for a day or so and reassess. Ramping back up too quickly can easily cause you to relapse and start the cycle over again.

Illness

Probably the second biggest reason that people get derailed from their plan, illness is one that usually carries with it the most potential for long-term damage. Many people attach a stigma of weakness to illness. Not wanting to admit to weakness, it's common for them to assume a state of denial about their illness. We have often heard, "Oh, I'm not sick. It's just a cold. I can train through it." This denial can result in that simple cold morphing into a sinus infection, upper-respiratory infection, or even pneumonia, all of which could take several weeks to fully recover from.

You have a cold because your body's natural defense mechanisms were not strong enough to keep this virus at bay. It is entirely likely that you are





Compulsion to Train

Letting a compulsion to train overpower the advice not to train when sick often results in the following pattern. See if this sounds familiar:

Day 1) Illness symptoms arise but you keep training normally for the first day.

Day 2) That night the symptoms worsen and so the next day you train lightly, thinking this will be enough to get over it.

Day 3) The symptoms are a bit worse so you take a day off.

Day 4) There is no change in the symptoms, but you get restless and go for a run.

Day 5) You are still not feeling great but some friends are going skiing, so you go too.

Day 6) You are feeling a bit worse, so you take the day off. You're going to shake this thing.

Day 7) Your symptoms are a little better, and you're back to training. Hard workout. Yahoo!

Day 8) Feeling really sick now, you take the day off.

Day 9) You have a hard time getting out of bed.

Day 10) Chills and fever.

Day 11) Still weak.

Day 12) A little better. No training.

Day 13) Got to do something! But had to cut the workout short—no energy!

Day 14) Back in bed with bronchial infection. On antibiotics now.

You can see how easily two weeks can slip away ineffectively trying to continue to train and ending up with an illness that will derail training for another ten days or so. What could have been brushed off in a week will now linger for the better part of a month. The result is that it will take a good deal of time to regain your previous strength.

sick because you are training too hard and/or not resting enough. While light-to-moderate training loads can boost natural immunity, heavy loads (high intensity and/or long duration) will suppress immunity, especially during the first hour immediately following the workout. Hence, the age-old parental admonition to change into dry clothes at the end of a hard workout and wear a hat in winter. Your body is especially vulnerable to infection during this window of time, and these extra precautions don't take much effort and really can pay off in keeping you healthy.

Our recommendation, after helping many athletes over many years, is that when the first signs of an illness show up, *stop training*. Your body and its limited

energy reserves are fully engaged in producing antibodies to fight the infection. Imposing an additional training load on a sick body will only further weaken your immune system and delay healing, as well as potentially open the door to additional infection. This simple but effective method guarantees that you'll be able to get back to effective training in the shortest amount of time.

AFTER THE ILLNESS

So what about after this cold or other minor illness? What is a good way to get back to regular training? You might still have some head congestion but your energy levels have improved. Even a common head cold is going to take two to three days to shake. Do not expect to jump right back to where you left off on that third or fourth day. Your aerobic system will have taken the biggest hit during your downtime. So, you need to return to training first with Zones 1–2 efforts.

To immediately engage in the same heavy training load you were doing before you got sick will usually be a letdown. Your perceived exertion will be higher as you are forced to rely on more anaerobic metabolism for energy. This will restrict the quality of the aerobic training you do during the first few days back. *We have found the simplest prescription for quickly regaining the flow in your training is to return to training with one day of easy training for every day you have been sick and away from training.* If you do this, you'll feel the improvement in aerobic capacity from day to day as it returns to normal. You can very gradually increase duration and intensity during these days. This method has worked well in many real-world applications. In most cases of minor illness, if you use this approach, within a week you will feel like your old self again.

Planned Breaks in Training

If you have the chance to anticipate an upcoming break in your schedule, it is possible to boost the training load for a short period, say a week, going into the business trip or family visit that's disrupting your training. Then, use that planned break as a recovery period, trying to get in just a few short, easy workouts. This sort of planned break following an overload period can be very beneficial to boost you to a higher level in the following days and weeks.

Try to incorporate this break into your plan a few weeks ahead of time so you can build into that hard week. This can remove the anxiety that often comes when missing training. You've planned for it, and you are using this break while visiting the parents or traveling for business as a chance to rebuild before tackling the next training block.

ATHLETE STORIES

Reframing Adversity

Jared Campbell

In my early years training for the Hardrock 100, I would joke with friends that anything difficult or unusually demanding was “good training for Hardrock.” A night run with a failed headlamp: good visual-acuity training. A long run after a stressful workweek without much

sleep: endocrine system training. Running out of food and entering depletion mode: an opportunity to condition the body to tap into fat reserves. Lack of traction on icy or slippery trails: proprioception honing. Post-holing in knee-deep snow: resistance training.



Jared Campbell running down from the Grant-Swamp Pass during the early stages of the 2012 Hardrock 100, San Juan Mountains, Colorado. Photo: Fredrik Marmsater

To be a mountain athlete is to be comfortable with rapidly shifting situations. I don't mean just having the latest waterproof jacket or headlamp; I mean being able to mentally deal with the wild ride that often accompanies mountain projects, especially as they grow in time and scale. A seasoned mountain athlete becomes skilled at working through and gracefully absorbing any flavor of obstacle or challenge—logistical, equipment related, nutritional, climatic. To that end, I have found that a regimen of intentional “adversity training” can add a valuable tool to the mountain athlete's quiver—and even provide a mental reframing approach that has applicability beyond just sport.

In 2005, I shattered my calcaneus (heel bone) in a climbing accident. I was told that I'd never run again and that it was time to take up a different sport. Six screws and one plate later I was on the mend and hell-bent on continuing my bipedal activities. The post-surgery time frame represented a fantastic opportunity for both mountain crutching and upper-body strength training. Turns out I loved my forearm crutches, and I went on to summit many of my favorite peaks. I even “crutched” a half marathon.

Six years later, I spent the winter of 2011 to 2012 training for my first Barkley Marathons, a wild 130-plus-mile footrace in eastern Tennessee with arduous routefinding, notoriously bad weather, 60,000 feet of vertical ascent, and a seemingly endless list of things that could go wrong. Since the race's inception in 1986, only ten people had successfully completed it, a finishing rate of less than 2 percent. At the time, I had enough experience to confidently construct a suitable physical training protocol. What I really needed was to be mentally ready.

Training in January and February in the Wasatch Mountains for a footrace was problematic, as more intelligent people were on skis. Given what I knew

about Barkley, I wanted to be completely comfortable operating in the worst conditions. I intentionally sought out the nastiest weather. If a storm was forecast to hit in the morning and clear in the evening, I would run in the morning. Given the infamous bushwhacking of the Barkley, I would eschew trails and vector up the most direct lines. If I found myself groveling through scrub oak, or scratching through manzanita bushes, I became thrilled with the opportunity I had been given for proper Barkley simulation.

I finished the Barkley in 2012 and went back to also finish in 2014 and 2016, becoming the race's first and only three-time finisher.

I have spent years consciously reframing normally bad situations into positive ones, an alternative response protocol that has enabled me to finish many of the toughest ultramarathon and mountain objectives. More importantly, however, it has positively influenced my perspective on nonrunning aspects of life as well. When confronted with dramatic or stressful life events, I find I am able to remain calm and measured. In every challenge there is something to be learned and a way to be a better person for it. In athletics as well as in life, find the silver lining and keep moving forward. ■

Jared Campbell is a Utah-based adventure athlete who is drawn to combining and applying multiple skills to accomplish seemingly impossible projects, from high-alpine linkups to multi-modality routes in the desert. In his teens and twenties, a healthy obsession with rock climbing took him across the globe. This led to bigger mountain linkups and, eventually, ultrarunning. He has completed more than thirty 100-plus-mile ultramarathons; he is the youngest ten-time finisher of the Hardrock 100 and the only three-time finisher of the Barkley Marathons.

Overtraining Syndrome (OTS)

Athletes who try to extract their maximum potential will be in a delicate balancing act, precariously close to the physiological edge between supreme fitness and utter physical collapse. The higher the level of the athlete, the narrower that edge will become, and the harder it is to balance there. Increasing fitness comes only as the result of increasing the training stress on the body. The elusive goal of every elite athlete is to find the proper balance between training and recovery. Not enough training stress and you will fail to achieve your potential for this training cycle, perhaps by a few percentage points. On the other hand, too much stress and you risk falling off the overtraining cliff. Doing so can mean that you will fall short of your potential by a huge margin. Balancing training loads with the additional stressors of life such as family obligations, school, or a demanding job all complicate the job of maintaining this critical balance. Ignore these normal life stresses at your peril.

Overtraining is a medical issue. In its simplest form, it manifests as an injury. But an insidious and demoralizing potential effect of overdoing structured endurance training is the failure of the body to adapt to the training stimuli. When in an overtrained state, you become less fit even though you are training at the same or higher level than before. In most endurance sports, overtraining is responsible for more failed athletic goals and it shortens more careers than any other factor besides injury. Some athletes struggle with it with several years of stagnation before throwing in the towel.

Endurance athletes find themselves at the highest risk for overtraining due to the high energy demands involved in their daily training regimen. These can exceed 6,000 kcal/day during very heavy training. Unfortunately, most athletes don't intuitively recognize overtraining. In many cases, it takes so long for the athlete to acknowledge overtraining that by the time it is properly diagnosed, it is usually too late for anything but the most extreme measures.

NOTES FROM KÍLIAN

In 2007, I overtrained. I did a big winter with thirty races and then I started running right away. The first indication was that I was not recovering at all. I was super-tired. Since I felt bad for a few weeks, I did a blood test to see if I had

an iron deficiency. Sure enough, I had super-low iron levels. So I rested completely for two full weeks. After that I started training again very, very easily. I think I caught it early, so I did not need to take much time off. Which was good.

Over- Versus Undertrained

It is far better to be slightly undertrained than to be even a little overtrained. This is an important training axiom to keep in mind. If you are undertrained and well rested, you can always draw a bit deeper from the willpower well in competition. However, if you are overtrained and tired, no amount of willpower can conjure strength and endurance from an empty well.

Motivation Versus Compulsion

Everyone involved in high-level athletics is driven to excel. Most endurance athletes fall squarely into the type A personality camp. These people typically adhere to the “more is better” training philosophy. After all, as a general rule, more endurance training *is* better. Willpower, perseverance, or drive—whatever you want to call the fire that burns hot inside your soul—is, without a doubt,



Fig. 4.1 Overtraining

Repeated training sessions without sufficient recovery time can easily lead to a long-term decline in fitness. This condition can take months to recover from if not recognized and corrected early.

the most important attribute you have as an endurance athlete. It can also be your worst enemy when it clouds your judgment and becomes a merciless taskmaster. Overtraining is not merely getting stiff and sore from a hard workout or needing a couple of extra days to recover from a particularly demanding race. In the exercise science world, that sort of short-term fatigue is called overreaching and is a normal part of the training cycle. When chronic overreaching develops into overtraining, a physiological bridge has been crossed that can be difficult to recross unless the athlete takes drastic action.

ATHLETE STORIES

A Lesson Learned the Hard Way: Training Isn't Competition

Marc Pinsach Rubirola

I have been training for ski-mountaineering and trail-running races since I was sixteen years old. Now I'm twenty-nine, and I have the background to analyze both the right and wrong decisions I have made about my training programs along the way.

One of the biggest issues with my training was that for a long time I didn't carry it out according to my personal needs and limitations. During these years, I had the opportunity to share training, races, and travels with Kilian Jornet. We are friends, and lived together while we raced and trained with the national team, and we grew both as athletes and people alongside each other. It was a pleasure and a great motivation to learn many things with him.

However, there was also a problem: I wasn't taking my time to grow up, to put some space between him and me, to find my own way in training. Consequently, my performance didn't increase.

Everybody knows that Kilian is a talent with exceptional physical gifts and technique, as well as unparalleled psychological abilities for training and racing. But his way to be fit was dangerous for someone like me, and sometimes I followed his example in my training programs. Kilian was always one or two steps above me, and his influence normally helped me to improve my performance level. But it also led to my being overtrained.

Learning to accept this performance gap—these differences between my body and Kilian's body—took time, some seasons off, and no small amount of psychological struggle. It nearly drove me into a burnout situation.

One of the things I thought I needed to do was to train more every year: more hours, more kilometers, more vertical meters. Until I was twenty-three, this

worked, and my performance grew according to the increase in training load. At that time, I was training 700 hours/year and doing 190,000 vertical meters/season in ski mountaineering. When I tried to increase the load to 850 training hours/year and 220,000 vertical meters/season, I ended up overtrained for four seasons.

It didn't help that my endurance pace during long training was too fast (Zone 2 and above), especially when I trained with other athletes. This pace caused me too much fatigue, so that by the time competition season started, my body was already under excessive strain. I accumulated even more fatigue by doing a lot of long intervals at aerobic pace, for example, three ten-to-fifteen-minute intervals at Zones 3–4. This meant I was spending too great a proportion of those training sessions at a fast pace, while at the same time failing to target competition pace. You are always up to Zone 4 in ski mountaineering or skyrunning races, usually racing for one-and-a-half to two hours.

These mistakes were rooted in psychological and emotional processes. My team (trainers, doctors, physiotherapists, sponsors, friends) alerted me to the risks of following this kind of training, and I understood. Nevertheless, I didn't pay them much attention because I had internalized the idea that in order to be in the top tier, I needed to train more and more—that “no pain, no gain” was the only way to get the best results.

The 2018 European Championships in the Vertical Race were held in Sicily on Mount Etna on a very foggy day. You couldn't see the other competitors and you didn't know if you needed to push to catch someone or to push not to be caught. The conditions reminded everyone that the best way to race was to concentrate on your body, mind, and technique. Italy. Photo: Jesus Dyañez



I also needed to feel fit, fast, and strong in each workout session and throughout the whole year. I didn't know how to rest and stay home without physical activity, and I strove for good marks and constant improvement in every training session. I treated each session like a race; I was always needing to prove to myself that I was strong and fit.

Maybe you can get good results in the short term this way, but my psychological and emotional configuration ultimately led to bad results, demotivation, and a lot of frustration and tiredness. In the end, I discovered that all of these wrong decisions were related to low self-esteem—an inability to trust in my personal way to get ready for race season.

It took time to restructure my mental approach and be competitive again. I had to accept that you can't increase training more than 10 percent between two consecutive seasons. I started being more gradual with my training, working my way back to the top slowly and over a period of years. I determined that my body can't train more than 700 hours/year and 200,000 vertical meters/season in ski mountaineering, and I now do the bulk of my training in Zones 1-2, at a very easy pace—the one I feel I could hold for hours without fatigue. And when I train with other people, I go my own pace, even if I end up running along with the group some meters in front of me. Training isn't competition day!

When it comes to intervals, I do repetitions at maximum speed for thirty seconds to three minutes. The most important aspect in this kind of training is that the speed is always maximum; if you can't do it at your

maximum, it's better to go home and recover in order to be ready for a new session.

You shouldn't train like your favorite athlete or your friends, because there aren't two identical people, and workouts don't produce the same effect in two different bodies. The most important thing is to find what your body needs regardless of what the other people around you are doing. I had to learn to control my emotions in order to be self-confident in doing my own training. I began to focus on myself in isolation, to bring to my body what it—and only it—needs to stimulate an increase in performance.

I now understand that in endurance sports, it takes a lot of time for the body to reach peak physiological development and, consequently, be primed for competition. During training sessions, you build what you will be in the future on race day. ■

Marc Pinsach Rubirola was born near the Pyrenees in Catalonia in 1989. When he was eighteen, he moved to Font Romeu in the French Pyrenees to study sports science and dedicate himself more fully to ski mountaineering and trail running. He now competes in around thirty races each year. Among his achievements are a skimo world championship title in the espoirs (under twenty-three) category (2011); fourth place in the Pierra Menta (2018) and the Patrouille des Glaciers (2010); third place in the Mezzalama Trophy (2011); a top five finish at the 2015 World Championships; and many top ten finishes in World Cup events.



A bird's-eye view of a competitor during the Vertical Crepa Neigra race, Val di Fassa, Italy.
Photo: Federico Modica

Early Signs and the Common Response

Overtraining is not your garden-variety fatigue that all endurance athletes learn to live with. When you step over this threshold, you are entering a medical condition that is poorly understood, multifaceted, and very difficult to diagnose except by those who have experience with it. Your family doctor is likely to call you a hypochondriac if you present the typical symptoms of overtraining. Even a moderately overtrained endurance athlete is still the picture of health and energy compared to the typical patient. Only in the latest stages of OTS will you present typical illness symptoms that are treatable with medical care.

OTS represents a breakdown of your body's natural adaptation processes. The endocrine and nervous system no longer behave as you have come to expect them to. What was once a light training load, because your body could adapt to it, even in a short-term fatigue state, is now an intolerable and destructive stress. You have to completely recalibrate your concept of *hard* and *easy*. Most of us are unprepared for this dramatic shift and unwilling to accept that we can go so quickly from feeling like an athlete to feeling like an invalid.



Rod Bien runs the 'Ōhana Trail near his hometown of Kailua on the island of O'ahu in Hawai'i.
Photo: Liz Barney

The first and most common sign of the early stages of overtraining is repeated shortcomings in performance accompanied by a feeling of flatness or low energy. The coach's or athlete's typical assumption is that this performance decrease is caused by a lack of fitness. After all, you were feeling *so* good just a couple of days before that it can't possibly be overtraining.

The common response when performance drops, especially by self-coached athletes, is to immediately add more training of some sort. The thought being that the training program lacks some key ingredient, and if they just add more of this or that, their lost fitness will return. After they increase their training with no improvement—and usually a further decrease in performance—they continue to seek solutions that involve more training. Thus begins a dangerous downward spiral that can ultimately lead to more than just physical problems. You have invested a huge amount of time, energy, and resources in your physical preparation. In many ways, your sport and your performance in it may define your persona. When you keep coming up short in training and races and suffer with constant deep fatigue, it *will* have dramatic and costly effects on your motivation and confidence.

Previous page: Anna Frost runs long up the northern side of Sermilik Fjord, East Greenland.
Photo: Kelvin Trautman

Overtraining is more easily recognizable if you are doing training on measured courses or using measured paces and see your times drop off. It's much harder to recognize in mountain sports where the track is different every day. This is one reason that we advise having benchmark workouts you can do to check for progress or regression in your program.

Symptoms

The earliest effects of overtraining impact the sympathetic nervous system. It starts with the immediate effect of raising the heart rate for submaximum levels of exertion compared to your normal heart rates. You just don't have your normal pep and vigor and feel flat in training. The levels of the stress hormone cortisol begin to stay elevated between training sessions (this takes a trip to the doctor to determine).

If nothing is done to mitigate the early overtraining, it can progress to a much more debilitating type involving the parasympathetic nervous system. When the parasympathetic nervous system gets involved, there are more negative hormonal effects and a *lowering* of the heart rate for all effort levels compared to normal rates.

When this occurs, watch for one or a number of these unpleasant symptoms:

- Persistent, deep fatigue
- Prolonged elevated cortisol levels (requires a blood test)
- Lowered testosterone levels (requires a blood test)
- Decreased heart rate variability (requires a special HR monitor or EKG)
- Irritability
- Depression
- Weight loss
- Amenorrhea (absence of menstruation)
- Insomnia
- Lowered libido
- Loss of enthusiasm and motivation

The profound state of deep neurological fatigue reached at this point necessitates, at the minimum, several weeks of complete rest and only then a gradual reintroduction of easy exercise. Sounds grim, right? It is a condition to be avoided at all costs. The best way to avoid this condition is to monitor your response to training by using the methods described earlier in this chapter. The cost of overtraining and underrecovering is so much worse than undertraining, it is far better to err on the conservative side.

Overuse Injury as Indicator of Overtraining

If a muscle is exposed to a new training load on a routine and consistent basis before adequate recovery and adaptation from the previous training session has occurred, at the very least, no fitness gains will be seen. More likely, however, is that the cumulative effect of this premature stress will cause small injuries that will weaken the muscle or tendon. The muscle/brain connection will then self-inhibit the ability to contract the muscle in an attempt to protect that muscle from further damage. A dedicated athlete can easily overcome this pain and suffering and manage to maintain the overuse, but soon a minor tear will turn into a major inflammatory cycle leading to overt injury and scarring. Chronic tendonitis or worse—full tendon or muscle rupture—can occur.

While these injuries are not necessarily an indication of overtraining as discussed above, they do point out that the adaptation we seek through training is not occurring. This lack of adaptation can be a good indicator that you are on a collision course with overtraining if some remedial steps are not begun.

Take these warning signs for what they are: your body's red flags to get your attention that things are amiss. Rather than get caught up in numbers and times, etc., pay attention to the bigger picture of your body's responses to the training.

Many endurance athletes get a little overtrained at one time or another and hopefully only once. The dramatic wasting effects of overtraining need to be treated with respect when training loads become high.

Periodization

The concept of periodizing sports training into discrete blocks with distinct training aims first arose in the former USSR and gained widespread recognition in the West largely from the writings of Soviet sports scientists. It has been proposed and, in many cases, verified that athletes may achieve better results when their training is organized in a way that focuses attention on only a few physical qualities at a time. Note the earlier distinction between Capacity and Utilization Training methods, which is a form of periodization.

Not everyone uses a periodized training plan. Some very successful athletes adhere to a simpler structure of maintaining all of their athletic qualities all the time by including all different types of training in all phases of their yearly plan. To use this method with the best chances of success requires considerable self-awareness and an extensive training history. Those who fit that description probably don't need this book anyway, so we are going to lay out a basic periodized plan to help guide the rest. Like all training theory, it is not cast in stone and is open to wide interpretation and application based on the individual.

The general idea of periodization is to train for a goal many months away during a *macrocycle*. This can be an annual or seasonal cycle. Once the length of

the macrocycle is set, then it is divided into several *mesocycles*. The mesocycles are where the distinct training targets occur. Each mesocycle will stand upon the preceding ones and support the following ones, so their organization is critical to the overall periodization structure. The mesocycles, which last a few weeks to a few months, are themselves divided into *microcycles*, typically seven days in length. Within the microcycle, the daily training prescriptions are used to influence the development of the qualities deemed appropriate for the mesocycle.

As the athlete progresses through the macrocycle and builds the capacity in the qualities that make up the event, it is important that those qualities previously built are not allowed to diminish to any great extent. Thankfully, it takes much less training stimulus to maintain a capacity than it did to first build it.

Transition Period

This period is meant to prepare the athlete for training either after a break following a major event or full competition season or when the athlete is new to regimented training. Depending on the length of the break that has occurred, the fatigue (mental and physical) carried over from the season/event, and the athletic maturity (in this sport) of the individual, this period may be as short as two weeks for an old hand coming off a very successful season or as long as eight weeks for the newcomer or very worn-down veteran.

Base Period

This period is where the most crucial work of training is accomplished. It is where the athlete focuses on increasing the work capacity of each of the individual fundamental qualities that are needed for his or her event. This is where the base of training is developed. You need this foundation to be rock-solid if you hope to construct a skyscraper of fitness on top of it. During this period your fitness will be increasing, but this will be masked by a low level of fatigue. You'll be carrying this slight fatigue load from day to day as an essential part of the necessary training adaptation process. Because of this, you should not expect your best performance results in training or competitions during this period.

Precompetition Period

This is where the sexy training gets added into the base training. It is training that in most aspects mirrors the specific demands of the event you are preparing for. It combines most if not all of those fundamental qualities you just spent months developing. Its length is measured in weeks rather than the months of the Base Period. If you've done your base building well, this creates a fitness level that you have never reached before. Capacity maintenance and recovery workouts fit between more demanding Utilization Training sessions. Fatigue levels are still going to be high,

but with better recovery before and after the most important workouts to allow them to be more effective. Examples of workouts in this period might be:

- If preparation for a VK: 8x3 min uphill in Zone 4 with 4-min easy walk-down recovery.
- If preparation for a 100-mile race: back-to-back eight-to-ten-hour days in Zones 1 and 2, three to four weeks out.

Competition Period

You've built up an impressive bank account of work capacity over the preceding several months. Now comes the time to start spending that money. Whether this period involves competitions or just personal achievements that push your limits, the principles will be the same.

This period is a delicate balancing act of wild spending sprees (competitions wherein you are utilizing all the capacity you have) followed by recovery—and hopefully interspersed with some capacity-rebuilding blocks during protracted inter-event periods to allow you to restock that bank account a bit. Depending on the number of races/events you are planning on and the stress imposed by them, you may very well not have much time to do extensive training once the race season is upon you. If you have a long, busy racing season, it can be helpful to set aside a training block during the race season where there are at least two weeks between shorter races and four to six weeks between ultradistance events to allow for the necessary recovery and subsequent rebuilding of the fitness base.

Recall that building fitness requires an extended period of chronically fatiguing yourself in order to get the adaptive stimulus needed to produce meaningful results. Undertaking such a program during the competition period, unless you find yourself with a few weeks between comps, will have little positive effect and potentially bad effects on your results. Blocks of two to three weeks between races can allow you to return to some maintenance-level base training, but remember that you'll be heavily fatigued after your last race and need time to recover from that. When you factor in that you'll also need some tapering time before your next race, your three weeks between races may allow only ten days of capacity-building training.

ATHLETE STORIES

The Life Cycle of an Ultrarunner

Krissy Moehl

I walked in the door from yet another long, cold, wet run—one of many over a particularly long, cold, and dreary Pacific Northwest winter. I was an emotional wreck after weeks of solo miles in depressing weather, my body often verging on hypothermia by the time I tried to unlock my front door. It was early February, just one month before I was set to race the 100-mile distance in the 2018 UTMB Gaoligong. I could tell that if I continued the training plan I'd originally mapped out, I wasn't going to make it to the finish line.

A 100-mile race takes more than training mileage to finish. It takes a deeper level of psych and motivation, which the dreary winter had sucked right out of me.

I am not a patient person by nature, but I am a patient runner, and the more I pay attention, the more I can apply the lessons I've learned through running to life. In this instance, I did what I'd instruct any of my coaching clients to do: I backed off. I cut my load from eighty to ninety miles a week to thirty, and I focused instead on strength training and flexibility. I cooked healing foods. I snuggled up with my pup on the couch. I laughed at the rain pounding on my window.

I knew balance was what I needed to regain my psych, but it wasn't an easy sell. While it's one thing to tell a client to rein it in, it's an entirely different conversation when it comes to myself. There is a confidence that comes with running. To build that same confidence and trust while sitting on the couch takes prior experience of pushing too hard and then failing on race day, and prior experience of resting and then having the best race of your life.

I didn't have that degree of understanding or patience fifteen years ago, when my then-husband and

I traveled to France for our honeymoon and raced the first edition of the Ultra-Trail du Mont-Blanc. In 2003, only a few years into my trail-running career, I was full of curiosity and drive to see what was physically possible.

That summer, France experienced a massive heat wave. I was expecting similar sweltering conditions on race day, but the week before UTMB, a storm swept in with polar-opposite weather: sideways rain and frigid temperatures. Completely unprepared, I had to borrow clothes along the course from strangers; I layered in cotton shirts just because it was a layer. My IT band gave out around mile sixty, so I walked the final thirty-plus miles—all with my jacket cinched so that only my eyes, nose, and mouth were exposed. I even walked across the finish line. I won, but it was essentially by default. The second-place woman came in eight hours later, and the majority of the field—men and women—had stopped at either the 50K or 100K finish lines.

That race, which took a lot of grit and determination to finish, provided a first glimpse of how different the 100-mile experience is from any of the distances that fall under it. Fifteen years and many hundred-milers later, I'm intimately familiar with both the distance and the time line for recovery—which has lengthened as I've gotten older. It's tough knowing that these big efforts will take me away from doing what I love—running—for two months while my body recuperates.

During those four weeks of pared-down training in advance of the 2018 UTMB Gaoligong, I regained my excitement for the race. My footwear sponsor had taken care of all the travel logistics, and they'd agreed to pay for my boyfriend, DJ, to join me as

